

Discrete Structures

Lecture # 1

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FAST -- National University of Computer and Emerging
Sciences. CFD Campus

Do not worry about your difficulties in Mathematics. I can assure you, mine are still
greater!

-Albert Einstein-

About me

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Research interests:

- Machine Learning, Computer Vision, Remote Sensing, Hyperspectral Imaging, Wearable Computing.

Outline

- ❑ What is DS and when can it be used?
- ❑ Why study DS?
- ❑ Topics
- ❑ Grading
- ❑ Today's Lecture

Discrete Structures

□ Study of Discrete Objects

Consisting of Distinct Objects

□ Problems Solved Using Discrete Structures

- How many ways are there to choose a valid password?
- What's the probability of winning a lottery?
- How to encrypt a message?
- What is the shortest path b/w two cities?
- How to sort a list of integers?
- How to prove that an algorithm works correctly?
- .
- .
- .

Why Study Discrete Structures???

❑ Ability to understand and create mathematical arguments

❑ Gateway to more advanced courses

- Algorithms
- Database theory
- Automata theory
- Compiler theory
- Computer security
- Operating system

Topics we'll study

- ☐ Logic and Proofs
- ☐ Mathematical Induction
- ☐ Sequences and Recursion
- ☐ Set Theory
- ☐ Functions
- ☐ Relations
- ☐ Counting and Probability
- ☐ Graphs and Trees
- ☐ Machine Learning (Tentative)

Course organization

- **Class Schedule**

- **Lecture # 1:** Check the Timetable ☺
- **Lecture # 2:** Check the Timetable ☺

- Class will be conducted using Slides

- **Text Book**

- Susana Epp, *Discrete Mathematics with Applications*, (4th Edition)
- Kenneth H. Rosen, *Discrete Mathematics and Its Applications*, 4th Edition

Grading

- ☐ **Two One Hour Mid Tests (30%)**
- ☐ **Final Exam (40%)**
- ☐ **Assignments (04)/Quizzes (04) (16%)**
- ☐ **Project (10%)**
- ☐ **Class Participation (4%)**

Other Info

☐ **Cooperation Policy**

☐ **Feedback**

☐ **Bonus Points**

☐ **Cheating Policy**

Policies

- No re-take of quizzes in any case except that the leave is officially approved by department
- Mid-term retakes are to be governed as per department policy
- Office hours will be announced in a couple of weeks. It is preferred to contact within office hours. Most of the problems can be dealt via email or in class.
- Cheating in one assignment= ZERO in two assignments
- Cheating in Quiz = ZERO in two quizzes
- Cheating in Project= ZERO in project
- Cheating in mid-terms and final exams are to be dealt as per department rules.

Today's Lecture

☐ **Integers:**

- Arithmetic Properties
- Powers
- Divisibility
- Primes of Composite Numbers

☐ **Rational Numbers**

- Equivalent fractions
- Operating with fractions
- Decimals

☐ **Irrational Numbers**

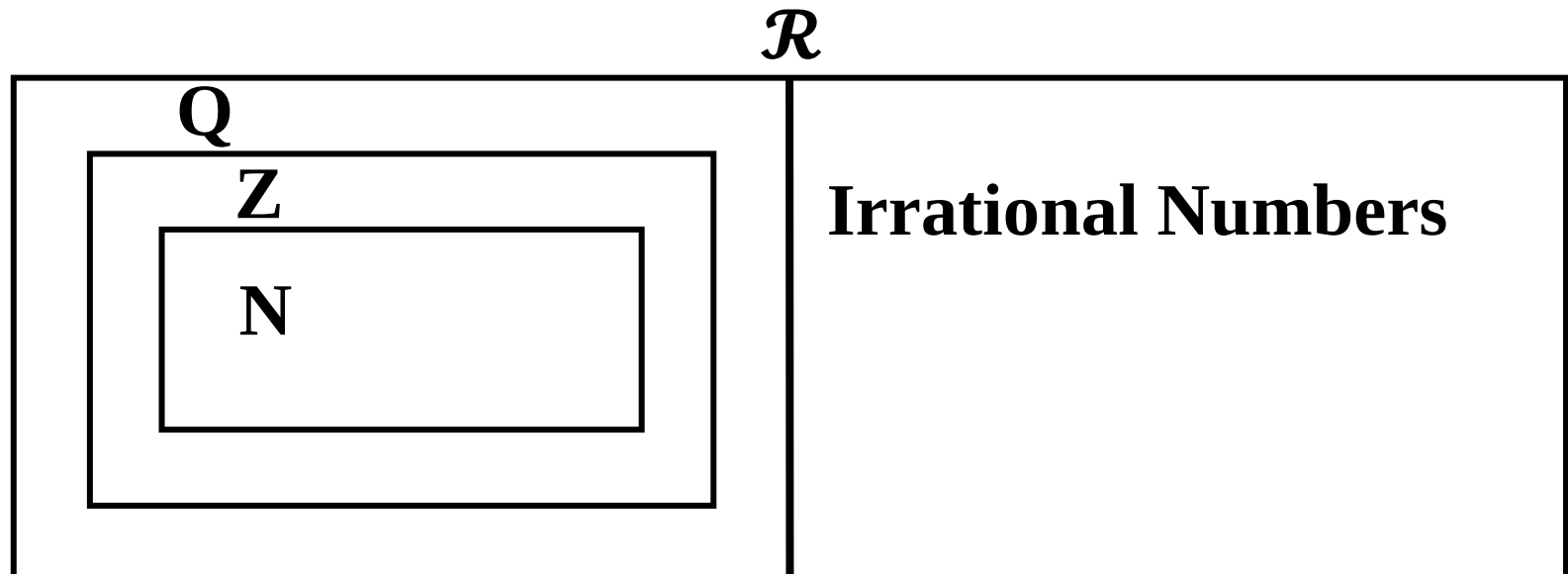
☐ **Real Numbers**

- Square roots
- N-th roots
- Logarithms
- Inequalities

☐ **Oder of Operations**

Numbers

- ❑ $\mathbf{N} = \{1, 2, 3, \dots\}$ The set of Natural Numbers
- ❑ $\mathbf{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ The set of Integers
- ❑ $\mathbf{Q} = \{p/q \mid p \in \mathbf{Z}, \text{ and } q \neq 0\}$ The set of rational numbers
- ❑ $\mathcal{R}$, the set of real number. e.g. Real Space



Integers

□ Simple rule of Addition

- For an integer a ,
- $0+a = a+0=a$
- $a+(-a)=0$, and $(-a)+a=0$
- $-a$ is the additive inverse of a .

We use “Minus a ” rather than “Negative a ”

Integers

❑ Rules of Addition

❑ Commutativity

- If a and b are integers, then
 - $a + b = b + a$

❑ Associativity

- If a , b and c are integers, then
 - $(a + b) + c = a + (b + c)$

Integers

□ Rules of Addition

- If $a + b = 0$, then $b = -a$ and $a = -b$

- Proof

$$a + b = 0$$

Add $-a$ to both sides

$$-a + a + b = 0 - a$$

$$0 + b = 0 - a$$

$$\mathbf{b = - a}$$

As desired.

Similarly we can find ---- $a = - b$

Integers

□ Rules of Addition

- If a , b are positive integers, then $a + b$ is also positive integer.
- If a , b are negative integers, then $a + b$ is also negative integer.
- If we have the relationship b/w three integers.
 - $a + b = c$

Then we can drive other relationships b/w them.

$$\mathbf{a = c - b} \qquad \mathbf{b = c - a}$$

□ Example: Solve for x .

$$\mathbf{x + 3 = 5}$$

$$\mathbf{x = 5 - 3}$$

$$\mathbf{x = 2}$$

Integers

□ Rules of Addition

- Cancellation rule for addition
 - If $a + b = a + c$, then $b = c$

Exercise:

Prove that if $a + b = a$, then $b = 0$?

Integers

□ Rules of Multiplication

□ Commutativity

- If a and b are integers, then
 - $a * b = b * a$

□ Associativity

- If a , b and c are integers, then
 - $(a * b) * c = a * (b * c)$
- For any integer a
 - $1 * a = a$ and $0 * a = 0$

Integers

□ Rules of Multiplication

□ Distributivity

- $a * (b + c) = a * b + a * c$
- $(b + c) * a = b * a + c * a$

Using all these properties

- $-1 * a = -a$
- $-(a * b) = (-a) * (b)$ or $-(a * b) = a * (-b)$
- $(-a) * (-b) = a * b$

Integers

□ Powers

- An exponent is used to indicate repeated multiplication.
- Tells how many times the base is used as a factor.
 - $a * a = a^2$
 - $a * a * a = a^3$

In general if n is a positive integer,

- $a^n = a * a * a \dots a$ (product is taken n times)

We say a^n is the n -th power of a .

If m, n are positive integers, then

- $a^{m+n} = a^m * a^n$

Integers

□ Powers

- $(a^m)^n = a^{m * n}$

Some important formulas

- $(a + b)^2 = a^2 + b^2 + 2ab$

- $(a - b)^2 = a^2 + b^2 - 2ab$

- $(a + b)(a - b) = a^2 - b^2$

Integers

□ Even and Odd integers

□ An even integer is an integer which can be written in the form **$2n$** for some integer **n**

- $2 = 2 * 1$
- $4 = 2 * 2$
- $6 = 2 * 3$

□ An odd integer is an integer that differs from an even integer by 1.

□ It can be written in the form **$2m \pm 1$** for some integer **m** .

- $1 = (2 * 1) - 1$
- $3 = (2 * 2) - 1$
- $7 = (2 * 3) + 1$

Integers

□ Theorem

- Let a, b be integers,
 - If a is even and b is also even, then $a + b$ is also even
 - If a is even and b is odd, then $a + b$ is odd
 - If a is odd and b is even, then $a + b$ is odd
 - If a is odd and b is also odd, then $a + b$ is also even

□ Exercise.

- Let's prove the Second statement ??

Integers

□ Divisibility

□ Given two integers a and b , with $a \neq 0$, we say that **a divides b** , or that **b is divisible by a** if there is an integer c , such that **$b = a * c$** .

□ Remember that every integer is divisible by 1 because we can always write

$$\bullet \quad n = 1 * n$$

□ Also, every positive integer is divisible by itself.

Rational Numbers

□ By a rational numbers, we mean a fraction as $\frac{m}{n}$,

where m and n are integers, $n \neq 0$.

- m is called numerator
- n is called denominator

□ Improper fraction

- m larger than or equal to n

□ Proper fraction

- m smaller than n

Rational Numbers

□ Equivalent Fractions

□ Two fractions that represent the same value.

- $\frac{1}{2} = \frac{2}{4}$

How can we know whether two fractions are equivalent?

□ Rule for cross-Multiplication

- Let m, n, r, s be integers and assume that $n \neq 0$ and $s \neq 0$. Then

- $\left(\frac{m}{n}\right) = \left(\frac{r}{s}\right)$, iff $m * s = r * n$

Rational Numbers

❑ Simplifying Fractions

❑ We can simplify four special fractions forms

❑ Fractions that have the same numerator and denominator.

$$\bullet \quad 1 = \frac{1}{1} = \frac{2}{2} = \frac{3}{3} = \frac{4}{4} = \dots$$

❑ Fractions that have a denominator of 1.

$$\bullet \quad \frac{5}{1} = 5, \quad \frac{24}{2} = 24, \quad \frac{-6}{1} = -6$$

❑ Fractions that have a numerator of 0.

$$\bullet \quad \frac{0}{8} = 0, \quad \frac{0}{71} = 0, \quad \frac{0}{-10} = 0$$

❑ Fractions that have a denominator of 0

$$\bullet \quad \frac{7}{0} = \infty, \quad \frac{-17}{0} = \infty, \quad (\infty = \text{Infinity} = \text{Undefined Value})$$

Rational Numbers

□ Simplifying Fractions

□ Cancellation Rule for Fractions

□ Let a be a non-zero integer. Let m , n be integers, and $n \neq 0$, then

$$\bullet \quad \frac{am}{an} = \frac{m}{n}$$

□ Proof: By applying the rule for cross-multiplication and using the associativity and commutativity laws.

Rational Numbers

□ Simplifying Fractions

□ A fraction is in **simplest form** when the numerator and denominator have no common factors (or divisors) other than 1.

□ Theorem:

□ “Any positive rational number has an expression as a fraction in the lowest form.”

Rational Numbers

□ Operating with Fractions

□ Addition (or Subtraction) with same denominator.

$$\bullet \quad \frac{a}{d} + \frac{b}{d} = \frac{a+b}{d} \quad \underline{\text{or}} \quad \frac{a}{d} - \frac{b}{d} = \frac{a-b}{d}$$

□ With different denominator:

$$\bullet \quad \frac{m}{n} + \frac{r}{s} = \frac{ms+rn}{ns} \quad \underline{\text{or}} \quad \frac{m}{n} - \frac{r}{s} = \frac{ms-rn}{ns}$$

□ Follows the same basic rules as addition of integers (commutativity and association)

Rational Numbers

□ Multiplication:

□ Let $a = \frac{m}{n}$

- Then for any positive integer k , such that

- $a^k = \left(\frac{m}{n}\right)^k = \frac{m^k}{n^k}$

- Follows the same basic rules as multiplication of integers.

Rational Numbers

□ Division:

□ If a is a rational number and $a \neq 0$, then there exists ($\exists$) a rational number, denoted by

- a^{-1} such that
- $a^{-1} * a = a * a^{-1} = 1$

□ Note that if $a = \frac{m}{n}$ then $a^{-1} = \frac{n}{m}$

□ a^{-1} is called the multiplicative inverse of a .

Rational Numbers

□ Decimals:

□ Finite decimals (and periodic) give us examples of rational numbers.

- $1.4 = \frac{14}{10}$
- $1.41 = \frac{141}{100}$
- $0.2 = \frac{1}{5}$
- $0.75 = \frac{3}{4}$
- $0.3333 = 0.\bar{3} = \frac{1}{3}$
- ...

Irrational Numbers

- ❑ A number that cannot be expressed as fraction of $\frac{p}{q}$ for any integers p and q .
- ❑ Have decimal expressions that neither terminate nor become periodic
 - $\sqrt{2} = 1.41421356237 \dots$
 - $\sqrt{3} = 1.73205080757 \dots$
 - $\pi = 3.14159265359 \dots$
 - $\dots$

Irrational Numbers

□ Is $\sqrt[2]{25}$ an irrational number?

- No!

- Because $\sqrt[2]{25} = \pm 5$

□ Is $\sqrt[2]{-1}$ an irrational number?

- No or Yes??? In both cases HOW?

Real Numbers

- ❑ Integers, Rational and Irrational Numbers are part of a larger system.
- ❑ Real Numbers can be described as all the numbers that consist of a decimal expansion, possibly infinite.

Real Numbers

□ Properties of Real Numbers:

□ Addition:

- $a + b = b + a$
- $a + (b + c) = (a + b) + c$
- For all ($\forall$) real numbers a , b , and c .

□ Multiplication

- $a * b = b * a$
- $a * (b * c) = (a * b) * c$
- $\forall$ real numbers a , b , c .

□ Also

- $a * (b + c) = a * b + a * c$
- $(b + c) * a = b * a + c * a$

Real Numbers

□ Absolute Value

□ The non-negative values of a real number without regard to its sign.

- $|a| = a$ for a positive a .
- $|a| = -a$ for a negative a
(in which case $-a$ is positive).
- $|0| = 0$

Real Numbers

□ Square Roots

□ If $a > 0$, then there exists ($\exists$) a number b such that (s.t).

- $b^2 = a$

□ N-th Roots

□ There exists a unique real number r such that

- $r^n = a$

It is called the n -th root of a , and is denoted by

- $a^{1/n}$ or $\sqrt[n]{a}$

Logarithms

- ❑ Can be seen as the reverse operation of the exponentiation.
- ❑ The logarithm of a number is the exponent to which another fixed value, the **Base** must be raised to produce that number.
 - $\log_{10}(10000) = 4$, because $10^4=10000$
 - $\log_2(16) = 4$, because $2^4=16$
 - $\log_3\left(\frac{1}{3}\right) = -1$, because $3^{-1}=\frac{1}{3}$

Logarithms

❑ Properties of Logarithms:

❑ Product:

- $\log_b(x * y) = \log_b(x) + \log_b(y)$

❑ Quotient:

- $\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$

❑ Power:

- $\log_b(x^p) = p * \log_b(x)$

❑ Change of Base:

- $\log_b(x) = \frac{\log_k(x)}{\log_k(b)}$

Inequalities

Symbol	Meaning	Example
$>$	Greater Than	$(X + 3) > 2$, for any X
$<$	Less Than	$(7X) < 28$, $X = \{ \dots, -2, -1, 0, 1, 2, 3 \}$
$\geq$	Greater Than or Equal	$5 \geq (X - 1)$, $X = \{ \dots, -2, -1, 0, 1, \dots, 5, 6 \}$
$\leq$	Less Than or Equal	$(2Y + 1) \leq 7$, $Y = \{ \dots, -2, -1, 0, 1, 2, 3 \}$

□ Let a, b, c be real numbers,

- If $a > b$ and $b > c$ then $a > c$. (Transitivity)
- If $a > b$ and $c > 0$ then $a * c > b * c$.
- If $a > b$ and $c < 0$ then $a * c < b * c$.